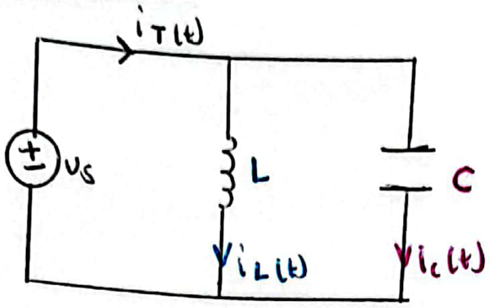
2024 EBN 111 A07

Question 1



Given:

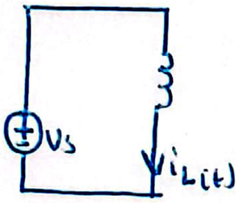
$V_s(t) = 2e^{-2t}$ V

$i_c(0) = 0A$

$i_L(0) = 0A$

*t (special param)

For $i_L(t)$



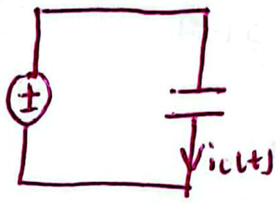
Given	Find	
V, L	i	integrate

$$i_L(t) = \left[\frac{1}{L}\int_{t_0}^{t_1} V_t\,dt\right] + i_{(0)}$$

$$= \left[\frac{1}{L}\int_0^t 2e^{-2t}\,dt\right] + 0$$

= (punch into calc)

For $i_c(t)$



Given	Find	
V, C	i	differentiate

$$i_c(t) = C\,\frac{dv}{dt}$$

$$= C \times \frac{d}{dt}\,2e^{-2t}\Big|_{t = \text{(given value)}}$$

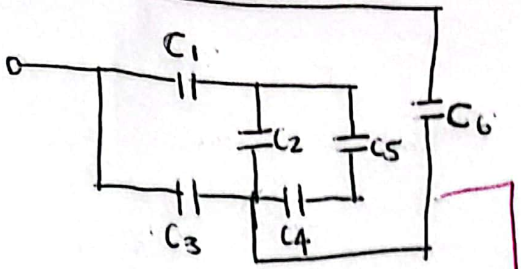
= (punch into calc)

$i_T(t) = i_L(t) + i_c(t)$

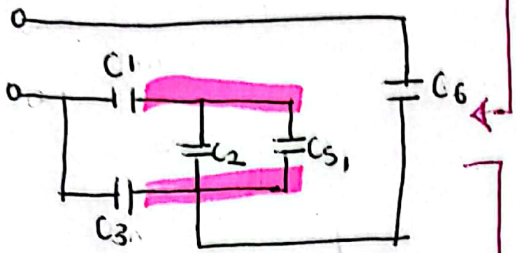
(be cautious of signs, do not ignore).

2)

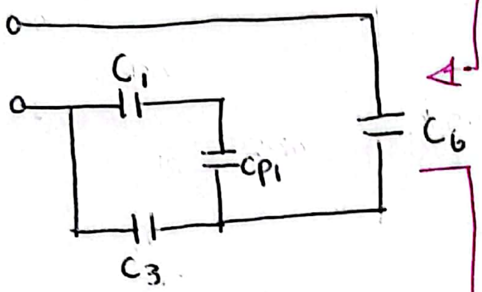
Question 2



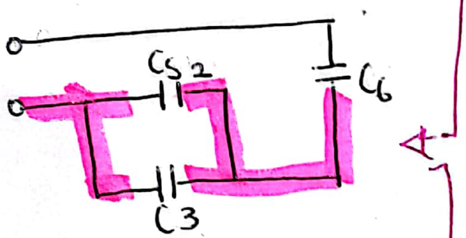
$$C_{s_1} = \left(\frac{1}{C_4} + \frac{1}{C_5}\right)^{-1}$$



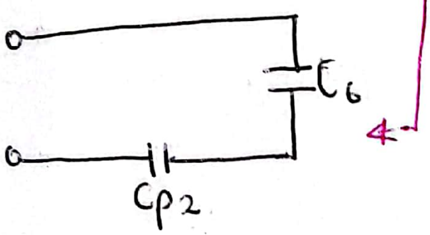
$$C_{p_1} = C_2 + C_{s_1}$$



$$C_{s_2} = C_1 + C_{p_1}$$



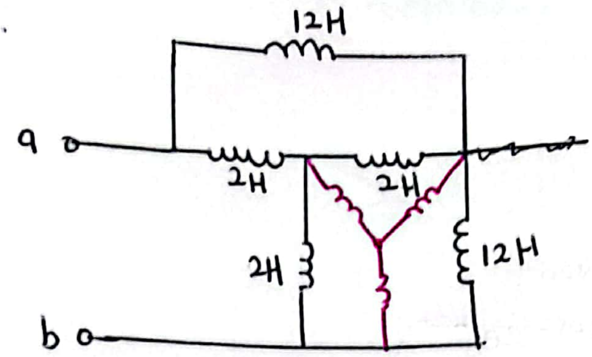
$$C_{p_2} = C_{s2} + C_3$$

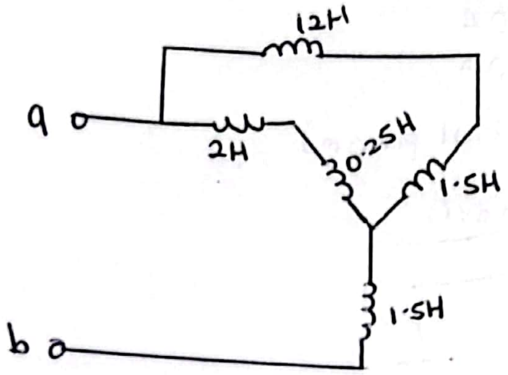


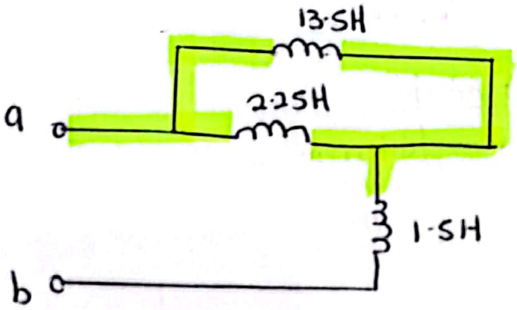
$$C_{eq} = \left(\frac{1}{C_{p_2}} + \frac{1}{C_6}\right)^{-1}$$

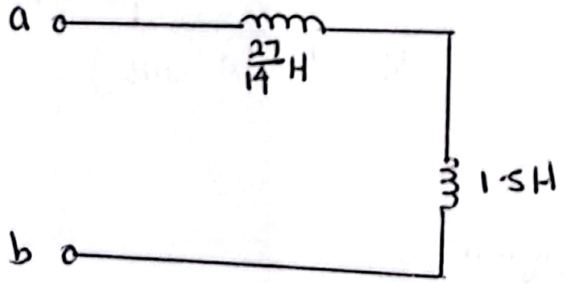
$=$ ______ F

3.









$$L_{ab} = \frac{27}{14} H + 1.5 H$$

$$= 3.43 H$$

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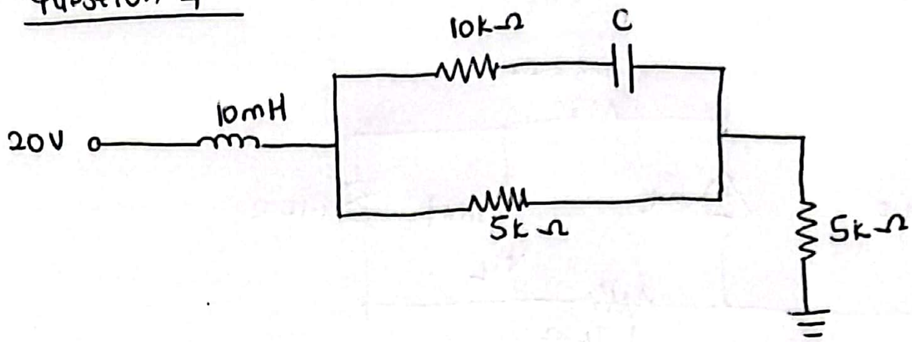
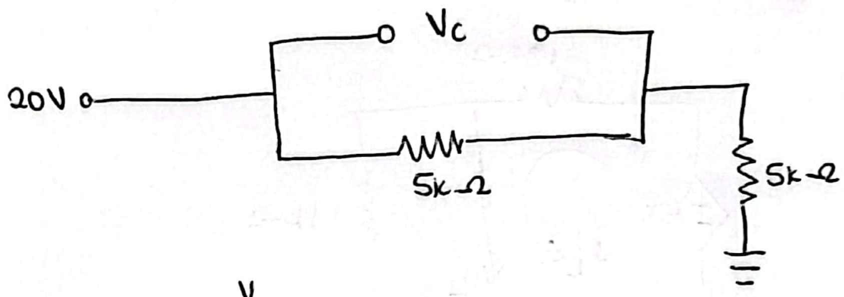
Question 4





$$I_{TOT} = \frac{V}{R_{TOT}}$$

$$= \frac{20}{5k+5k} = \frac{20}{10k} = 2 \times 10^{-3} A$$

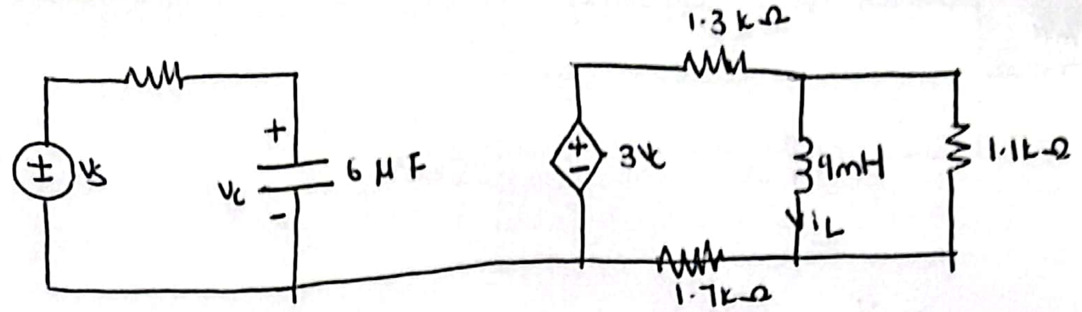
$$V_c = V_{5k} = (2 \times 10^{-3})(5k) = 10V$$

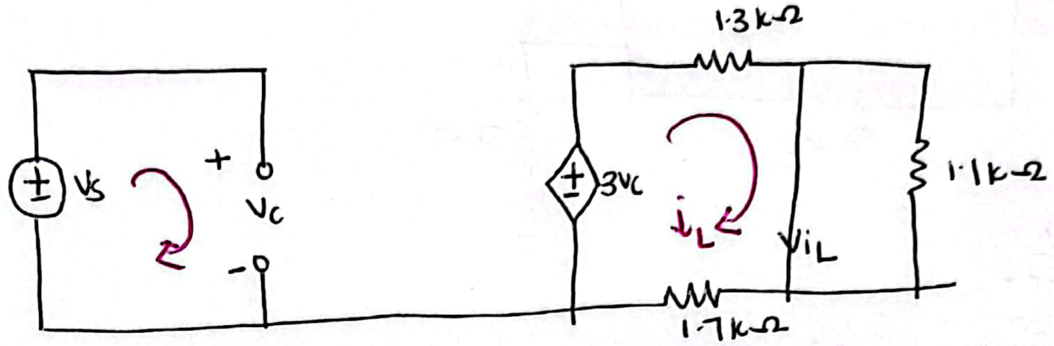
$$w_c = \frac{1}{2} C V_c^2$$

$$= \frac{1}{2} C (10)^2$$

$$= \_\_\_\_\_ J$$

Question 5





$-V_s + V_c = 0$

$V_C = V_S$

$1.7k\, i_L - 3V_s + 1.3k\, i_L = 0$

$\Rightarrow$ say $i_L = 0.05\text{ A}$

$W_L = \frac{1}{2} L i_L^2$

$= \frac{1}{2} L (0.05)^2$

$=$ ______ J

Question 6

$$Z = \frac{(3 - j4)(-3 + j4)}{\sqrt{4\angle A} + j\sin 30^\circ}$$

Into calc : $\frac{(3-4i)(-3+4i)}{(\sqrt{4} \angle \frac{A}{2}) + \sin 30^\circ i}$

you prolly gonna get your answer in rectangular form.
to put ~~it~~ in polar form $\Rightarrow$ "Shift", "2", ~~"3~~ "3"

On Excel : 1.23 $arg; 4.56 $ deg;